Chemistry

Unit 3

Practice Exam 1

This sample exam has been prepared as part of the Pearson suite of resources for the Year 12, Unit 3, ATAR Chemistry Course prescribed by the Western Australian School Curriculum and Standards Authority.

Time allowed

Reading time: 10 minutes Working time: 90 minutes

Materials required

An approved non-programmable calculator.

Chemistry Data Sheet. This may be downloaded from the WACE website.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of total exam
Section one: Multiple-choice	16	16	24	32	26
Section two: Short answer	9	9	36	49	40
Section three: Extended answer	3	3	30	43	34
				Total	**100**

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Section one: **Multiple-choice** 26% (32 marks)

This section has **16** questions. Answer **all** questions by circling the correct option.
Do not erase or use correct fluid/tape. Marks will not be deducted for incorrect answers.
No marks will be given if more than one answer is completed for any question.
Suggested working time: 24 minutes

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1 Consider the following equation.

$Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

Which one of the following statements is correct about this reaction?

a HCl is acting as a strong acid.

b $ZnCl_2$ is a conjugate base.

c H^+ is acting as an oxidising agent.

d Zn is acting as an oxidising agent.

2 What is a conjugate acid–base pair in the following reaction?

$HS^-(aq) + OH^-(aq) \rightarrow S^{2-}(aq) + H_2O(l)$

a $HS^-(aq)$ and $OH^-(aq)$

b $HS^-(aq)$ and $S^{2-}(aq)$

c $OH^-(aq)$ and $S^{2-}(aq)$

d $S^{2-}(aq)$ and $H_2O(l)$

3 Chlorous acid, $HClO_2$, has a K_a of 1.2×10^{-2} and nitrous acid, HNO_2, has a K_a of 4.0×10^{-4}.

Considering equal concentrations of each of these acids, which one of the following statements is correct?

a Chlorous acid is the stronger acid and its solution will have the higher pH.

b Nitrous acid is the stronger acid and its solution will have the higher pH.

c Chlorous acid is the stronger acid and its solution will have the lower pH.

d Nitrous acid is the stronger acid and its solution will have the lower pH.

4 Consider 100 mL of 0.10 mol L^{-1} solutions of each of these compounds: potassium carbonate, aluminium chloride and potassium chloride.

Which one of the following alternatives lists these solutions in order of increasing pH?

a potassium carbonate, potassium chloride, aluminium chloride

b aluminium chloride, potassium carbonate, potassium chloride

c potassium chloride, aluminium chloride, potassium carbonate

d aluminium chloride, potassium chloride, potassium carbonate

5 Beakers A and B both contain nitric acid. The pH of the acid in beaker A is 3 whereas the pH of the acid in beaker B is 1. From this information what can we deduce about the concentration of hydrogen ions in beaker A?

a It is three times that in beaker B.

b It is one-third of that in beaker B.

c It is one thousand times that in beaker B.

d It is one-hundredth of that in beaker B.

6 Which one of the following can act as either a Brønsted-Lowry acid or base in aqueous solutions?

a HS^-

b NH_4^+

c CO_3^{2-}

d CH_4

7 The first step in the commercial production of nitric acid involves the oxidation of ammonia into nitrogen(II) oxide according to the following equation.

$4NH_3(g) + 5O_2(g) \leftrightharpoons 4NO(g) + 6H_2O(g) \quad \Delta H < 0$

Which one of the following alternatives describes the effect of an increase in temperature at constant volume on:

I the value of the equilibrium constant, *K*, and

II the amount of NO(g) produced at equilibrium.

	Value of equilibrium constant	Amount of NO(g) produced at equilibrium
a	unchanged	increased
b	unchanged	decreased
c	increased	increased
d	decreased	decreased

8 A reaction vessel contains an equilibrium mixture of H_2(g), I_2(g) and HI(g) at 40°C. The equation for the reaction is given below.

$H_2(g) + I_2(g) \leftrightharpoons 2HI(g)$

The volume of the reaction vessel is decreased while the temperature is maintained at 40°C.

What effect will this change have on the amount of HI at equilibrium, the concentration of HI at equilibrium and the rate of the forward reaction?

	Effect on amount of HI at equilibrium	Effect on concentration of HI at equilibrium	Effect on the rate of the forward reaction
a	no change	no change	no change
b	no change	increased	no change
c	increased	increased	increased
d	no change	increased	increased

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9 An electrochemical cell consists of a Mn^{2+}(aq)/Mn(s) half-cell and a half-cell composed of a mixture of Sn^{2+}(aq) and Sn^{4+}(aq) with a $C_{(graphite)}$ electrode. The half-cells are joined by a potassium nitrate salt bridge.

It is found that electrons flowed through the connecting wire from the Mn electrode to the $C_{(graphite)}$ electrode.

Which one of the following is the strongest oxidising agent?

a Sn^{2+}

b Sn^{4+}

c Mn

d Mn^{2+}

10 A ring is to be covered with a thin layer of gold by electroplating. Where should the ring be attached in order to achieve this, and how does it act?

a negative terminal of the power supply and acts as the anode

b negative terminal of the power supply and acts as the cathode

c positive terminal of the power supply and acts as the anode

d positive terminal of the power supply and acts as the cathode

11 Which one of the following compounds contains phosphorus with a different oxidation number to the others?

a P_4O_6

b $POCl_3$

c H_3PO_4

d Na_2HPO_4

12 Which one of the following manganese-containing compounds is likely to be the strongest oxidising agent?

a K_2MnO_4

b $KMnO_4$

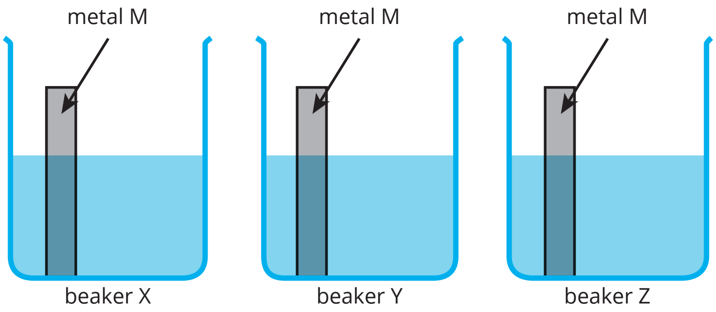
c Mn_2O_3

d MnO_2

Questions 13 and 14 refer to the following information.

Three beakers are set up at 25°C as shown below. Each beaker contains one of 1.0 mol L^{-1} $Pb(NO_3)_2$, 1.0 mol L^{-1} $Zn(NO_3)_2$ and 1.0 mol L^{-1} $Cu(NO_3)_2$ but not necessarily in this order.

A piece of unknown metal M is dropped in each beaker.



In beakers X and Y, a metal displacement reaction is observed. No visible reaction is noted in beaker Z.

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13 From this information, what can we deduce about the standard reduction potential of

$M^{2+}(aq) + 2e^- \leftrightharpoons M(s)$?

- **a** It is higher than 0.34 V.
- **b** It is between −0.13 V and 0.34 V.
- **c** It is between −0.76 V and −0.13 V.
- **d** It is less than −0.76 V.

14 Which one of the following species is the strongest reducing agent?

- **a** metal M
- **b** M^{2+} ions
- **c** Zn metal
- **d** Cu^{2+} ions

15 Which of the following reactions occur spontaneously?

I $Br_2(aq) + 2I^-(aq) \rightarrow 2Br^-(aq) + I_2(aq)$

II $Br_2(aq) + 2Cl^-(aq) \rightarrow 2Br^-(aq) + Cl_2(g)$

- **a** I only
- **b** II only
- **c** I and II
- **d** neither I nor II

16 Ammonia, NH_3, acts as a weak base in water.

$NH_3(aq) + H_2O(l) \leftrightharpoons NH_4^+(aq) + OH^-(aq)$

Two drops of concentrated KOH are added to the above equilibrium at constant temperature. As a result, what occurs at the new equilibrium?

- **a** The pH of the solution will be higher.
- **b** The concentration of $OH^-(aq)$ will be lower.
- **c** The concentration of $NH_3^+(aq)$ will be lower.
- **d** The value of the equilibrium constant will be lower.

End of section one

Section two: **Short answer** 40% (49 marks)

This section has **9** questions. Answer **all** questions. Write your answers in the space provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them.

Suggested working time: 36 minutes

Question 17 **(8 marks)**

Consider the following chemical equations and answer the questions below.

$Pb^{2+}(aq) + 2Cl^-(aq) \rightarrow PbCl_2(s)$

$Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$

$HCOOH(aq) + H_2O(l) \rightarrow HCOO^-(aq) + H_3O^+(aq)$

$2NO_2(g) \rightarrow N_2O_4(g)$

$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$

a **i** Write the equations for **2** reactions which can be classified as acid–base reactions.

ii Explain why each of your selected reactions is considered to be an acid–base reaction.

(4 marks)

Equation	Explanation

b **i** Write the equations for **2** reactions which can be classified as oxidation/reduction reactions.

ii Explain why each of your selected reactions is considered to be a redox reaction.

(4 marks)

Equation	Explanation

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Question 18 **(5 marks)**

A solution of 100 mL of 0.10 mol L^{-1} ethanoic acid is added to 100 mL of 0.100 mol L^{-1} sodium ethanoate. The mixture reaches equilibrium.

a Write a balanced equation for this equilibrium. (1 mark)

b Explain why there is little change to the pH of this mixture with the addition of a small amount of the following. (3 marks)

i HCl

ii NaOH

c What name is given to a mixture that resists changes in pH when a small amount of acid or base is added? (1 mark)

Question 19 **(4 marks)**

Sodium hydrogencarbonate is the key ingredient of household baking soda. It contains the HCO_3^- ion, which is able to form both a conjugate acid and a conjugate base.

a Give an equation for the reaction of the HCO_3^- ion when it forms its conjugate acid and one when it forms its conjugate base. (2 marks)

HCO_3^- ion forming its conjugate acid

HCO_3^- ion forming its conjugate base

b The pH of a dilute solution of sodium hydrogen carbonate is approximately 8. Use this information to deduce which of the two reactions in part a occurs to the greater extent. Explain your reasoning. (2 marks)

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Question 20 **(5 marks)**

Beaker A contains 100 mL of 0.10 mol L^{-1} nitric acid, a strong acid, and beaker B contains 100 mL of 0.10 mol L^{-1} nitrous acid, a weaker acid. Both beakers are maintained at 25°C.

For each of the statements below, place a tick in the appropriate column to indicate if the value of the property indicated is higher for nitric acid, higher for nitrous acid or the same for both.

	Higher for nitric acid	**Higher for nitrous acid**	**Same for both**
pH			
concentration of hydronium ions			
amount of hydroxide ions			
K_a value			
K_w value			

Question 21 **(6 marks)**

An experiment was conducted to determine the concentration of a solution of hydrochloric acid. A 20.00 mL sample of sodium carbonate, of concentration 0.0150 mol L^{-1}, was pipetted into a conical flask and two drops of indicator added. Hydrochloric acid was then added from a burette until an end point was reached. The procedure was repeated until concordant results were obtained.

a Indicate what should be used to rinse the following pieces of apparatus used in this experiment. (3 marks)

	Should be rinsed with ...
conical flask	
pipette	
burette	

b What characteristic must a substance have to be useful as an acid–base indicator? (1 mark)

c For this experiment, identify one source of systematic error and one source of random error. (2 marks)

systematic error	
random error	

Question 22 **(3 marks)**

Complete the following table which relates the chemical equation of a reaction at equilibrium to the equilibrium law expression for the reaction.

Equation	**Equilibrium law expression**
$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$	
$Ag^+(aq) + Fe^{2+}(aq) \rightleftharpoons Ag(s) + Fe^{3+}(aq)$	
	$K = \frac{[NOCl]^2}{[NO]^2[Cl_2]}$

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Question 23 **(9 marks)**

In a closed vessel, hydrogen peroxide (H_2O_2), bromide ions, hydrogen ions and bromine are in equilibrium according to the following equation.

$H_2O_2(aq) + 2Br^-(aq) + 2H^+(aq) \rightleftharpoons Br_2(aq) + 2H_2O(l) \quad \Delta H < 0$

The colour of the equilibrium mixture is light orange.

a Which species in the equilibrium mixture is responsible for the orange colour? (1 mark)

____

b Predict whether the intensity of the orange colour in the equilibrium mixture would increase, decrease or remain the same if each of the following changes are made. Provide an explanation for your prediction in terms of any changes to the rates of the forward and reverse reactions and the position of equilibrium. (8 marks)

Change	**Intensity of orange colour of mixture increased/ decreased/the same**	**Explanation in terms of reaction rates and position of equilibrium**
The mixture is heated.		
A small amount of concentrated KBr solution is added.		
A small amount of concentrated KOH is added.		

Question 24 **(5 marks)**

The amount of $SO_2(g)$ in polluted air can be determined by bubbling the air through an excess of acidified $KMnO_4(aq)$. Two of the products of the reaction are $SO_4^{2-}(aq)$ and $Mn^{2+}(aq)$.

a Write the half equation for the oxidation reaction. (1 mark)

____

____

b Write the half equation for the reduction reaction. (1 mark)

____

____

c Deduce the overall ionic equation for the reaction. (1 mark)

____

____

d What is the oxidation number of sulfur in the $SO_4^{2-}(aq)$ ion? (1 mark)

____

____

e The amount of $SO_2(g)$ is then indirectly determined by back titrating the leftover $MnO_4^-(aq)$ with $H_2O_2(aq)$. Explain why there is no need to add an indicator for this titration. (1 mark)

____

____

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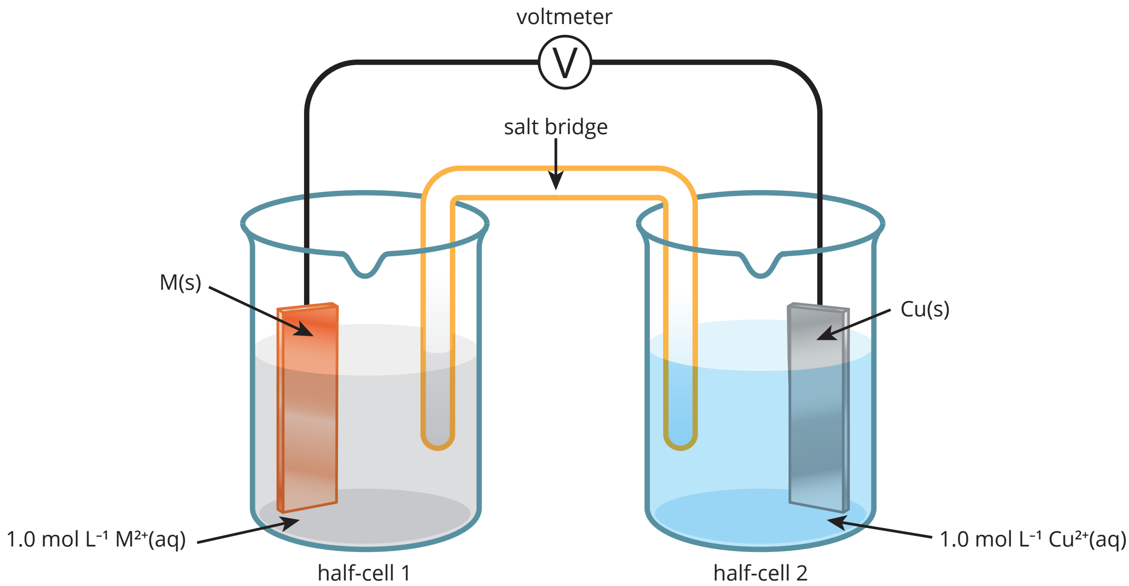
Question 25 **(4 marks)**

The following electrochemical cell is constructed.



Half-cell 1: an electrode of metal M in a 1.0 mol L^{-1} solution of M^{2+}(aq) ions

Half-cell 2: an electrode of copper metal in a 1.0 mol L^{-1} solution of Cu^{2+}(aq) ions

a After some time, the blue colour of the Cu^{2+}(aq) ions solution became lighter.

Use this information to deduce which species in this cell is the stronger reducing agent and explain your deduction. (2 marks)

b If the predicted cell potential for this cell is 1.52 V under standard conditions, what would be the standard reduction potential for the half-cell M^{2+}/M? (2 marks)

End of section two

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Section three: **Extended answer** 34% (43 marks)

This section has **3** questions. Answer **all** questions. Write your answers in the space provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them.

Suggested working time: 30 minutes

Question 26 **(15 marks)**

A solution of the strong acid hydrochloric acid, HCl, has a concentration of 0.010 mol L^{-1} and a volume of 20.00 mL.

a Calculate the following.

i concentration of OH^- ions in mol L^{-1} in the solution (1 mark)

____

____

ii pH of the solution (1 mark)

____

____

iii amount, in mol, of H^+ ions in the solution (1 mark)

____

____

b Compare each of the following solutions with 20.00 mL of 0.010 mol L^{-1} HCl with respect to pH and the amount of 0.010 mol L^{-1} NaOH it would require for neutralisation. Include a justification for your prediction. (6 marks)

	In comparison to 20.00 mL 0.010 mol L^{-1} HCl,	
	the pH of the solution is ...	**the amount of 0.010 mol L^{-1} NaOH the solution would need for neutralisation is ...**
40.00 mL 0.010 mol L^{-1} HCl	higher/the same/lower Justification:	higher/the same/lower Justification:
20.00 mL 0.010 mol L^{-1} of the weak acid HOCl	higher/the same/lower Justification:	higher/the same/lower Justification:
20.00 mL of 0.010 mol L^{-1} H_2SO_4	higher/the same/lower Justification:	higher/the same/lower Justification:

c In order to determine the accurate concentration of a solution of potassium hydroxide, 20.00 mL potassium hydroxide was placed in a conical flask and titrated against 0.100 mol L^{-1} HCl, which was placed in a burette.

Four titrations were performed and the volume of HCl titres is given below.

Titration	1	2	3	4
Volume of HCl-delivered mL	17.60	17.30	17.35	17.25

i Write a balanced ionic equation for the reaction. (1 mark)

ii Why was it necessary to perform 4 titrations? (1 mark)

iii Calculate the concentration of the KOH solution in mol L^{-1}. (2 marks)

iv The following indicators are available for the titration.

Indicator	**Colour of acid form**	**Colour of base form**	**pH range over which the colour changes**
methyl orange	pink	yellow	3.1–4.4
bromothymol blue	yellow	blue	6.0–7.6
phenol red	yellow	red	6.8–8.4
phenolphthalein	colourless	pink	8.3–10

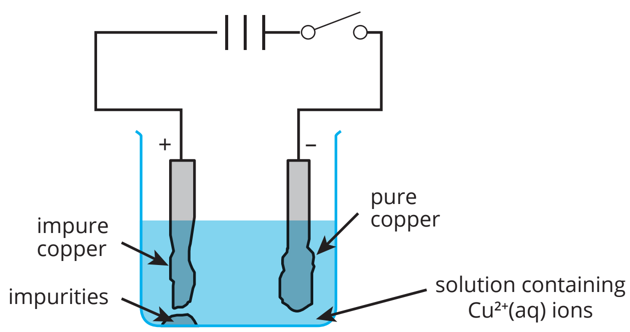
Student A says that either bromothymol blue or phenol red, but not the other two indicators, should be used in this titration. Student B, however, says any one of the above indicators would be suitable. Which student is correct and why? (2 marks)

Question 27 **(14 marks)**

Electrolytic cells are used in industry to extract metals such as aluminium from their ore as well as to refine metals such as copper.

a Electrolysis is used to purify copper for use in electricity wires. A particular sample of copper contains 1% impurities. These impurities consist of silver, gold, zinc and lead.

A diagram of the electrorefining cell is shown below.



i Write half equations for any reaction that occurs at the cathode and at the anode of the cell. (4 marks)

at cathode

at anode

ii What happens to the silver and gold impurities as the electrolysis proceeds? Why? (2 marks)

iii What happens to the zinc and lead impurities as the electrolysis proceeds? Why? (2 marks)

iv A student observed that the colour intensity of the electrolyte remains unchanged during the process. Explain. (1 mark)

v What mass of pure copper will be deposited at the negative electrode for every 1.0 mol of electrons that pass through the circuit? (2 marks)

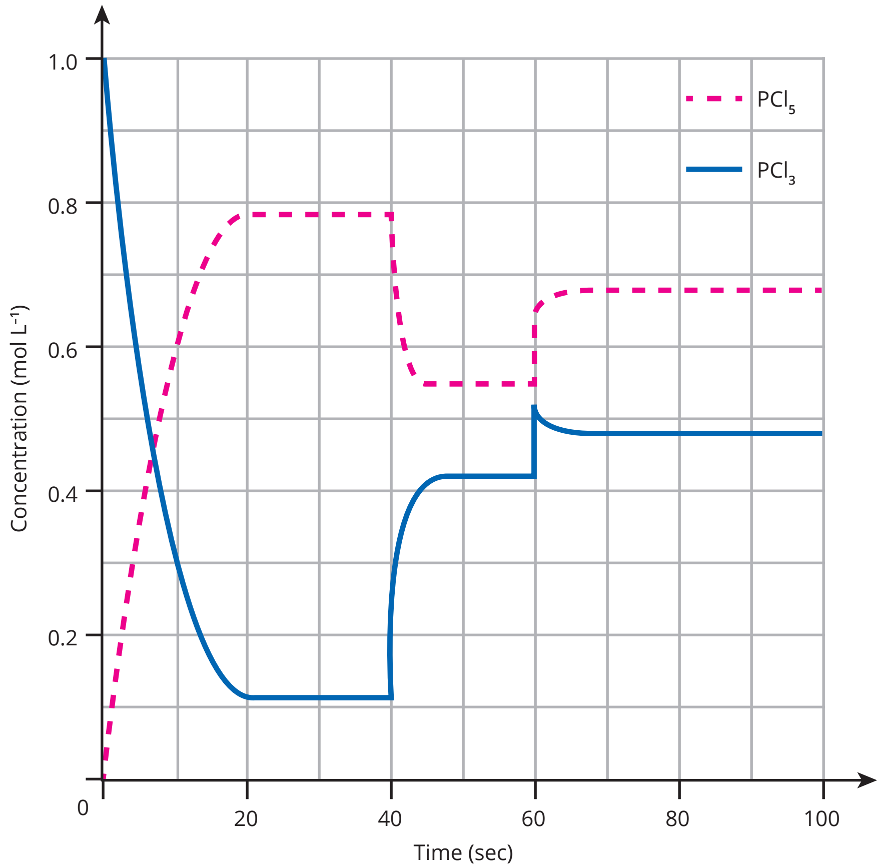
b Copper can be deposited by electrolysis of an aqueous solution. However, the electrolytic production of aluminium requires that water be excluded and that the electric current be passed through a molten mixture of aluminium compounds. Explain. (3 marks)

Question 28 **(14 marks)**

One mole of PCl_3(g) and one mole of Cl_2(g) are placed in an empty container. After 20 seconds, the following equilibrium is established.

$PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$

The graph below is a plot of the concentration of PCl_3(g) and PCl_5(g) against time. At the 40-second and 60-second marks, the equilibrium system was disrupted in some manner and the graph indicates the resultant changes in the concentrations of PCl_3(g) and PCl_5(g).



a Write the equilibrium law expression for this equilibrium system. (1 mark)

b Between 0 and 40 seconds, the temperature was maintained at about 225°C. Use the information provided by the graph for that period to deduce if the value of the equilibrium constant at that temperature is greater or smaller than 1 and provide an explanation for your deduction. (2 marks)

c At the 40-second mark, the temperature of the system was suddenly increased to about 325°C at constant volume and a new equilibrium was established. The temperature was then maintained at 325°C for the rest of the experiment.

i Use the information provided by the graph to deduce if the reaction as written is exothermic or endothermic. Provide an explanation for your deduction. (2 marks)

ii Will the value of the equilibrium constant at 325°C be larger, smaller or the same as at 225°C? Explain. (2 marks)

d At the 60-second mark, a different change was imposed on the equilibrium.

i Suggest what that change could be. (1 mark)

ii Explain why this resulted in a subsequent increase in the concentration of $PCl_5(g)$ but a decrease in the concentration of $PCl_3(g)$. (1 mark)

iii Explain why the concentrations of both $PCl_5(g)$ and $PCl_3(g)$ are higher at the 70-second mark than at the 50-second mark. (1 mark)

e **On the graph itself**, show how the concentration of $PCl_3(g)$ would change over the period from 0 to 40 seconds, if the experiment were repeated but with a suitable catalyst added. (1 mark)

f It is found that at 25°C, it takes about 2 hours to produce 100 g $PCl_5(g)$ by this reaction. However, at 600°C it takes less than 1 second to produce that same amount of $PCl_5(g)$. Use collision theory, to explain why this is so. (3 marks)

End of questions

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